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Super-slow nonlinear hysteresis loop "tracking" for managing energy intake in the forced Duffing oscillator

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ABSTRACT

Hysteresis loops are ubiquitous in harmonically (and not only) forced nonlinear oscillators. These loops result due to the well-known nonlinear bi-stability phenomenon, i.e., the co-existence of steady state responses, with the initial conditions dictating the attraction of a specific orbit by either of these steady states. This introduces an element of uncertainty in the forced dynamics. Hence, if one targets the excitation of a higher-amplitude co-existing steady state solution, e.g., to maximize the energy input into the forced oscillator, this sensitivity on initial conditions introduces uncertainty. Moreover, the role of such hysteresis loops in terms of nonlinear physics is not entirely clear; this contrasts with similar hysteresis loops appearing in, e.g., in cyclically loaded viscoelastic materials, which denote the energy dissipated per excitation cycle. Here a methodology is presented for removing the uncertainty of the forced dynamics on initial conditions, and, in the process, for better understanding and exploiting nonlinear hysteresis loops. Considering the forced Duffing oscillator, as an example, harmonic excitations with super-slowly modulated amplitudes are considered as a means of "tracking" the hysteresis loop during a super-slow cycle of the applied modulated force. For either single or repetitive cycles of super-slow force modulations one may tune the modulations to predictively maximize or minimize the energy intake into the forced oscillator depending on which regions of the hysteresis loop are tracked. Importantly, computational studies indicate that the forced dynamics become independent of the specific initial conditions, thus removing the uncertainty in the response due to bi-stability. These findings elucidate the physical significance of nonlinear hysteresis in terms of energy transfer in forced oscillators, and are applicable to a broad class of forced dynamical systems exhibiting nonlinear hysteresis.

1. Introduction

In nonlinear dynamics, major attention is commonly paid to periodic excitations and periodic response regimes, starting from basic resonances in quasilinear systems [1,2] and up to advanced analytic and numeric explorations of nonlinear normal modes [3,4]. This preference is well-justified. First of all, the periodic regimes are crucial and desirable in a plethora of applications. Besides, they yield powerful analytical and numerical methods for computation, based on harmonic analysis (Fourier series) or fixed-point approaches. Last but not the least, a stability of the periodic regimes can be analyzed based of the convenient and precise framework of Floquet theory [2].

However, in many problems, the assumption of exact periodicity can be too restrictive or even somewhat misleading. For instance, in integrable systems the generic solution is quasiperiodic, since for periodicity, one requires that all frequencies are commensurable, with zero measure [5,6]. On the other hand, in non-integrable Hamiltonian systems stable periodic orbits are generically surrounded by families of quasiperiodic (and chaotic) orbits [7,8]. Lastly, in the forced-damped nonlinear systems, quasiperiodic responses can be readily found as attractors even in the case of perfectly periodic forcing [9,10].

Moreover, a well-known feature of nonlinear systems is the possible co-existence of stable solutions (attractors). Indeed, the possible multiplicity of realizable response regimes under a given excitation is a salient feature of nonlinear dynamical systems [1,2], and can be demonstrated even in the simplest models, e.g., the harmonically forced Duffing oscillator. Such multiplicity of attractors introduces an uncertainty in the forced dynamics, since it is the specific choice of initial conditions

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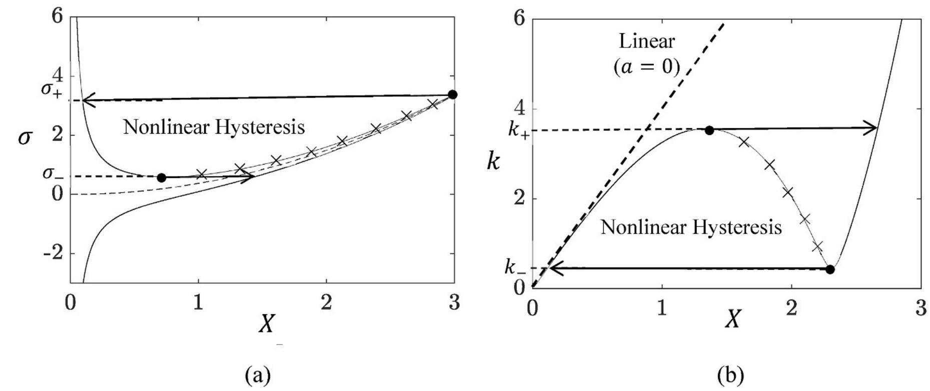


Fig. 1. Approximate steady state of the forced oscillator (1) (slow invariant manifold – SIM) for $\zeta = 0.1$ and $\alpha = 1$: (a) Response amplitude versus frequency detuning for fixed forcing amplitude $k = 0.6$, and (b) response amplitude versus forcing magnitude for fixed frequency detuning $\sigma = 2.0$ (for comparison, the corresponding linear invariant manifold is depicted by dashed line); nonlinear hysteresis loops are being formed in the bi-stability regions of the two plots, bounded by the fold points which are denoted by bullets (●), while the unstable branches are denoted by crosses (×).

that dictates the attractor that will eventually attract the transient dynamics. This effect, well-known as *multi-stability*, is since each of the multiple possible co-existing stable response regimes has its own basin of attraction, usually with complicated fractal boundaries, making the control of the precise response a challenging task, and introducing significant uncertainty in the engineering design of these systems.

Intuitively, one could assume that quasiperiodic forcing may lead to even more complicated dynamics, magnifying even further the multi-stability effect by introducing additional possible stable responses. The main purpose of this work is to demonstrate precisely the opposite effect, namely that *quasiperiodic forcing with appropriate parameters can remove completely the multi-stability effect* as well as the associated uncertainty of the response with respect to the specific choice of the initial conditions. Section 2 presents the analytic foundations of this finding and the underpinning of the followed methodology on multi-scale analysis. Section 3 presents a robustness study, that is numerical verifications and parametric studies verifying the elimination of the sensitivity of the forced dynamics on the initial conditions. Finally, section 4 is devoted to conclusions and discussion of possible implications and applications of the methodology of this work to broader areas.

2. Response of the duffing oscillator with amplitude-modulated harmonic force

Consider the normalized Duffing oscillator with weak hardening nonlinearity forced by a weak harmonic excitation,

$$\ddot{x}(t) + x(t) + 2\varepsilon\zeta\dot{x}(t) + \varepsilon\alpha x^3(t) = \varepsilon k \cos(1+\varepsilon\sigma)t \tag{1}$$

where, $0 < \varepsilon \ll 1$ is a bookkeeping small parameter, and ζ, α and k are the viscous damping ratio, the coefficient of nonlinearity, and the forcing amplitude, respectively; lastly, σ indicates a frequency detuning forcing from frequency unity. All variables apart from ε are assumed to the $O(1)$ quantities, and real constants, unless otherwise specified.

Equation (1) defines a classical problem in nonlinear dynamics which is well studied [1]; accordingly, in the following we provide briefly the known approximate solution of (1), since it will be the basis for our understanding of the response of the same system subject to modulated harmonic excitation. Since (1) is a quasi-linear dynamical system, i.e., a nonlinear perturbation of a linear oscillator, it can be analyzed by any standard method, e.g., averaging [2,11], multiple scales [1], or complexification-averaging [12]. The fast time scale can be averaged out, yielding the equation on the slow time scale (εt). Fixed points of this slow-flow equation correspond to the steady-state responses of the initial system and satisfy the following well-known relationship:

$$Z\left[4\zeta^2 + \left(2\sigma - \frac{3\alpha Z}{4}\right)^2\right] - k^2 = 0 \tag{2}$$

In (2), $Z \equiv X^2$ and $X = |\dot{x} + jx|$ represents the modulus of the complex amplitude of the asymptotically approximated steady state response, and $j = (-1)^{1/2}$. In the literature, the curve defined by Equation (2) is referred to as slow invariant manifold (SIM) [9] or critical manifold (Fr). The term SIM stems from the fact that this manifold is a set of fixed, i.e., invariant points with respect to the slow flow.

Differentiating (2) with respect to σ, one finds the points $(\sigma_\pm, Z_\pm)$ at the fold points:

$$\frac{\partial Z}{\partial \sigma} \to \infty \Rightarrow \quad Z_\pm = \frac{8}{9\alpha}\left[2\sigma_\pm \pm \left(\sigma_\pm^2 - 3\zeta^2\right)^{1/2}\right] \tag{3}$$

Substituting back into (2) yields the following expression for the frequency detuning coefficients at the fold points:

$$\pm\frac{4}{3}\left(\sigma_\pm^2 - 3\zeta^2\right)^{3/2} = 4\sigma_\pm\left(\frac{\sigma_\pm^2}{3} + 3\zeta^2\right) - \frac{27}{16}\alpha k^2 \tag{4}$$

This equation is not solvable analytically, but some conclusions are available. Indeed, for given (a, ζ) there exists a minimum excitation amplitude, for which one obtains *bi-stability*, i.e., co-existence of two stable steady state solutions:

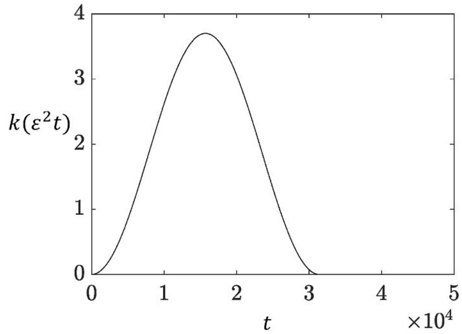
$$k_{min} = \frac{16}{3^{5/4}}\left(\frac{\zeta^3}{\alpha}\right)^{1/2} \tag{5}$$

Then, for the case of weak damping (namely, $\zeta \ll \sigma$), one can asymptotically approximate the detuning values at the fold points as:

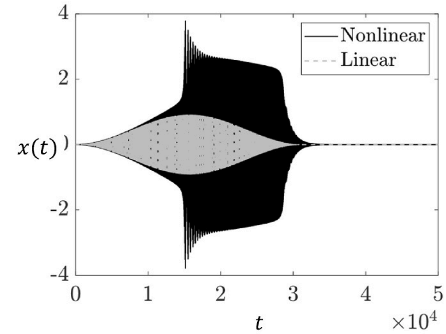
$$\sigma_+ = \frac{3\alpha k^2}{32\zeta^2} + O(1), \sigma_- = \sigma_*\left(1 - \frac{3\zeta^2}{4\sigma_*}\right) + O(\zeta^4), \sigma_* = \left(\frac{81\alpha k^2}{128}\right)^{1/3} \tag{6}$$

The SIM for parameters $k = 0.6, \zeta = 0.1$ and $\alpha = 1$ is presented in Fig. 1a for varying frequency detuning σ. For these numerical values, the two fold points that bound in frequency the region of bi-stability are computed as, $\sigma_+ = 3.376$ and $\sigma_+ = 0.598$. This is the traditional way to present the approximate steady state response of (1) and to depict the nonlinear hysteresis loop that is formed due to the co-existence of two stable branches of the SIM (with a connecting unstable branch formed in between). It is possible to prove that for almost any initial condition of the slow flow (besides the stable manifold of the unstable branch, which, however, has only mathematical and not practical significance) the system will be attracted to one of the stable branches [9].

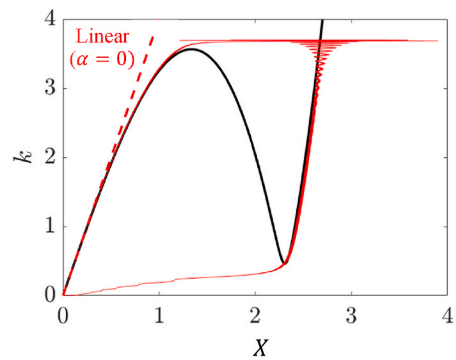
It is possible, however, to represent the SIM (2) in a different way, that is, by fixing the detuning (say, $\sigma = 2.0$) and, instead, allowing the forcing magnitude k to vary. The corresponding approximate steady



(a)



(b)



(c)

Fig. 2. Response of oscillator (9) for zero initial conditions subject to the super-slow cycle of modulated forcing magnitude depicted in (a): Strongly modulated nonlinear and linear responses (b), and projection of the modulated response amplitude onto the SIM (c); the super-slow tracking of the hysteresis loop in this case is evident.

state representation (SIM) is depicted in Fig. 1b, where again a hysteresis loop is formed due to bi-stability. In this case the upper and lower fold points occur at the following bounding values of the forcing magnitude:

$$k_{\pm} = \frac{8}{9\alpha^{1/2}}\left[\sigma\left(\sigma^2 + 9\zeta^2\right) \pm \left(\sigma^2 - 3\zeta^2\right)^{3/2}\right]^{1/2} \tag{7}$$

For $\alpha = 1.0, \sigma = 2.0$ and $\zeta = 0.1,$ these extreme values are given by $k_{+} \sim 3.565$ and $k_{-} \sim 0.462$. In Fig. 1b, the two stable branches of the SIM, which are connected by an unstable branch, form a nonlinear hysteresis loop (which is different in topology compared to the traditional hysteresis loop in Fig. 1a). The structure of the SIM in Fig. 1b resembles analogous SIMs reported in systems of nonlinear coupled oscillators [9,13].

The SIM points are stationary only at the slow time scale εt. As it is well-known, at even slower time scales, the system can "track" the SIM, e.g., due to nonlinear resonant interactions. For some settings, this tracking results in relaxation oscillations, that is, in oscillation cycles composed of slow parts which are connected by fast transitions. In Fig. 1b we depict also for comparison the linear SIM corresponding to $\alpha = 0$, showing the *drastic* change in the topology of the slow dynamics induced by the (*weak*) hardening stiffness nonlinearity. This provides ample motivation to explore the possibility of tracking the SIM in Fig. 1b (i.e., the nonlinear hysteresis loop) by means of adding an extra component of super-slow dynamics in the forcing, i.e., forcing terms that are much slower than the slow dynamics governing the SIM.

Accordingly, we consider the following *single super-slow cycle of forcing amplitude modulation*,

$$k\left(\varepsilon^2 t\right) = \left[k_0 + k_1\, sin^2\left(\varepsilon^2 t\right)\right]\, \left[1 - U\left(t - \frac{\pi}{\varepsilon^2}\right)\right] \tag{8}$$

where k_0 and k_1 are real constants, and $U(\bullet)$ is the Heaviside function. Note that alternative forms of such super-slow modulations may be considered, and (8) is used only as an example. Moreover, *repetitive super-slow cycles of force modulation* may be considered, as shown later. In turn, the equation of the forced Duffing oscillator (1) takes the form,

$$\ddot{x}(t) + x(t) + 2\varepsilon\zeta\dot{x}(t) + \varepsilon\alpha x^3(t) = \varepsilon k\left(\varepsilon^2 t\right)\, cos(1 + \varepsilon\sigma)t \tag{9}$$

where we take $\varepsilon = 0.01$, i.e., sufficiently small so to be in accordance with the assumptions of the asymptotic theory. The rationale behind this modification is that now the magnitude of the force (but not its frequency detuning σ) is function of the super-slow time scale $\varepsilon^2 t$. It follows that, in accordance to asymptotic theory (time scale separation), variables depending on the fast and slow time scales, t and εt, respectively, can be considered as constants in the super-slow time scale $\varepsilon^2 t$, so the super-slow variations of these variables should undergo (independent) modulations. We note that the independence between responses at different time scales, although approximate, is a basic assumption of standard asymptotic theory, e.g., in the method of multiple-scales or averaging [1], provided that the nonlinearity is weak – which holds if $\varepsilon \ll 1$ in (9).

We now show that depending on the choice of the modulation parameters k_0 and k_1, it will be possible to track different branches of the nonlinear hysteresis loop, which, in turn, yields to corresponding variation of energy intake in the oscillator by the modulated excitation. We start by assuming the previous values of the system parameters and setting $k_0 = 0$ and $k_1 = 3.7$, i.e., at a value slightly above k_{+} – see (7). The super-slow modulation of the force is depicted in Fig. 2a. The resulting strongly modulated response of (9) for initial conditions $x(0) = \dot{x}(0) = 0$ is depicted in Fig. 2b, with the corresponding linear response to the modulated force also shown for comparison. The projection of the super-slow variation of the modulus of the amplitude $X\left(\varepsilon^2 t\right) = \left[x^2\left(\varepsilon^2 t\right) + \dot{x}^2\left(\varepsilon^2 t\right)\right]^{1/2}$ on the SIM is shown in Fig. 2c, where again the linear projection is shown with dashed line. We note that *the entire*

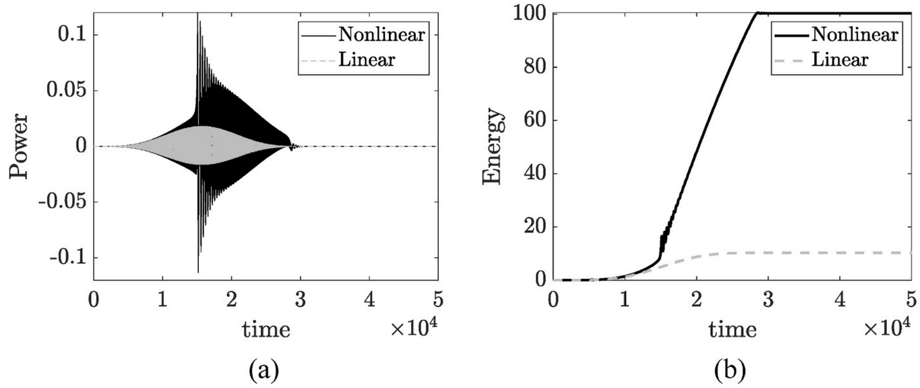


Fig. 3. Power (a) and energy intake (b) (normalized units) in the oscillator (9) with zero initial conditions, and subject to the super-slow cycle of modulated force depicted in Fig. 2a; the corresponding results for the linear system ($\alpha = 0$) are shown for comparison.

nonlinear hysteresis loop is tracked due to the selected force modulation, which results in drastically increased nonlinear responses compared to the linear case.

We emphasize the occurrence of strong overshoot effects that are realized during the (fast) transition from the low-amplitude stable branch of the SIM to the higher-amplitude stable branch. As mentioned before, at the slow time scale the flow is attracted to the SIM. A remarkable feature of the forced dynamics is that *the depicted super-slow cycle of the response is robust to the initial conditions*, so the tracking of the hysteresis loop and the excitation of the higher-amplitude stable branch are independent of the specific initial conditions. The geometric reason for this insensitivity with respect to the initial conditions is that the super-slow flow encompasses the whole hysteretic cycle, regardless to which branch it is attracted in the beginning of the tracking.

It is of interest to compute the resulting energy intake to the system by the modulated force and compare it to the linear system. This comparison is depicted in Fig. 3, where the input power and the corresponding energy intake (which is defined as the overall work of the external force within a given time interval, or, equivently, by the energy dissipated by the viscous damper of the Dffing oscillator during the same time interval) are depicted for the super-slow modulated cycle of the forcing magnitude of Fig. 2a. The results confirm drastic increase in energy intake in the nonlinear system (a nearly ten-fold increase) even though the stiffness nonlinearity is weak ($\varepsilon = 0.01$). Clearly, this increase is exclusively due to the tracking of the upper stable branch of the hystsresis loop, which is not possible in the linear system. Moreover, as shown in the next section these results are robust to changes in initial conditions, since the tracking of the slow invariant manifold (hysteresis loop) is proven to be insensitive to the specific initial conditions; this removes the well known uncertainty in the responses of harmonically forced nonlinear oscillators due to bi-stability.

Moreover, complete circulation of the SIM loop by the super-slow flow is a sufficient, but not necessary condition to remove the uncertainity of the response with respect to the initial conditions. Indeed, one can arrange the parameters of the quasiperiodic forcing in a way that the super-slow flow will either stay on the prescribed stable SIM branch from the very beginning, or will jump to it from the other branch, while never transitioning (jumping) back, and all this *irrespective of the initial conditions*. Thus, it is feasible to *predictibly manage the energy intake in the oscillator (9) by suitable selecting the force modulation parameters, without being restricted by the bi-stability of the forced nonlinear dynamics*. To demonstrate this, we consider repetitive cycles of force modulation in the form,

$$k(\varepsilon^2 t) = k_0 + k_1 \sin^2(\varepsilon^2 t) \tag{10}$$

And demonstrate that by appropriately choosing the constants k_0 and k_1, we may increase or decrease the energy intake by selecting which regions of the SIM are tracked by the forced dynamics. For the following results, the parameters $\varepsilon = 0.01, \zeta = 0.3, \alpha = 1.0$ and $\sigma = 2.0$ are used, zero initial conditions are imposed, and two different repetitive force modulations (8) are considered, namely:

$$k_I(\varepsilon^2 t) = 3.1 + 0.6 \sin^2(\varepsilon^2 t) \text{ and } k_{II}(\varepsilon^2 t) = 3.1 + 0.7 \sin^2(\varepsilon^2 t) \tag{11}$$

The selection of the parameters of the force modulations (11) is governed by relation (7), which in this case yields $k_+ = 3.64, k_- = 1.38$.

We note that athough the two modulations differ only slightly by the magnitudes of their (super-slowly) oscillating components, the resulting responses of oscillator (9) are drsatically different since different regions of the hystsresis loop are tracked. In Figs. 4 and 5, the responses due to modulations $k_I(\varepsilon^2 t)$ and $k_{II}(\varepsilon^2 t)$ are shown, respectively. Refering to

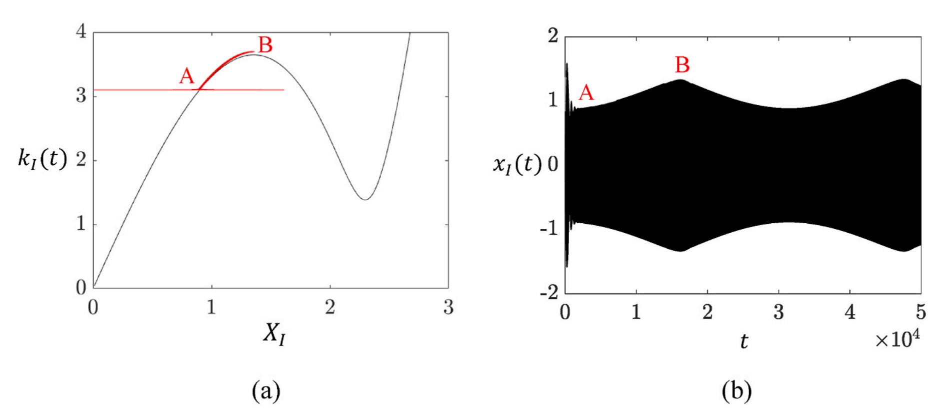


Fig. 4. Response of (9) due to the modulated forcing magnitude $k_I(\varepsilon^2 t)$: (a) Projection of the forced dynamics on the SIM, and (b) corresponding super-slow modulated response $x_I(\varepsilon^2 t)$.

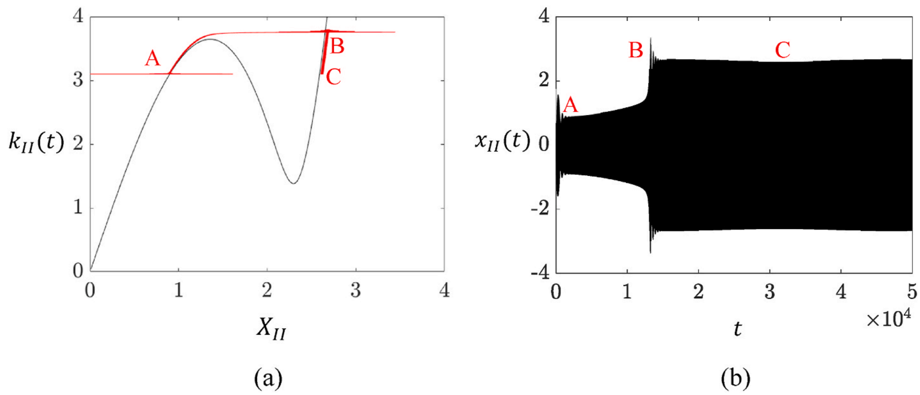


Fig. 5. Response of (9) due to the modulated forcing magnitude $k_{II}(\varepsilon^2 t)$: (a) Projection of the forced dynamics on the SIM, and (b) corresponding super-slow modulated response $x_{II}(\varepsilon^2 t)$; note the drastically increased response compared to Fig. 4b, caused by a slight increase in the modulation of the forcing magnitude.

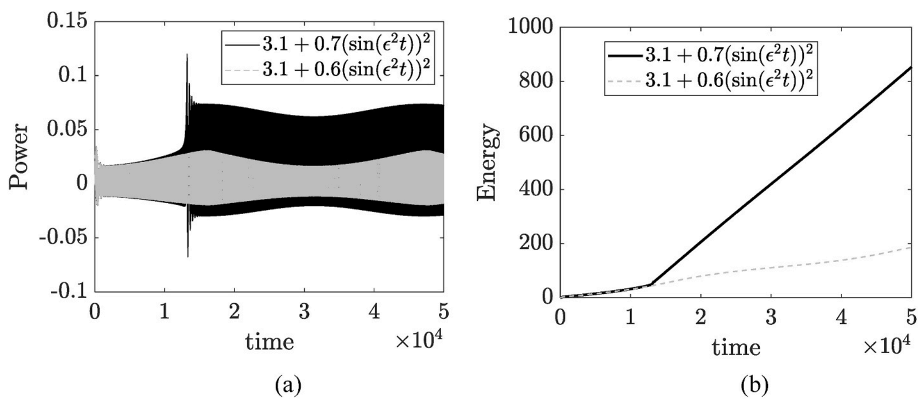


Fig. 6. Power input (a) and energy intake (b) in the oscillator (9) for applied harmonic forcing with magnitude modulations $k_I(\varepsilon^2 t)$ and $k_{II}(\varepsilon^2 t)$; these results correspond to the responses presented in Figs. 4 and 5.

Fig. 4, the force modulation $k_I(\varepsilon^2 t)$ oscillates between points A and B on the lower-amplitude stable branch of the SIM, so the response undergoes a low-amplitude, super-slow modulatiom since the forced dynamics does not make the fast transition (jump) to the higher-amplitude stable branch f the SIM (as the oscillating component in $k_I(\varepsilon^2 t)$ is not sufficiently large).

A slight increase in the oscilalting component of $k_{II}(\varepsilon^2 t)$ yields a drastically different response (see Fig. 5), since in that case there is a transition to the higher-amplitude stable branch of the SIM, which, in turn, results in drastically increased response amplitudes. It is interesting to note that *after the transition, the dynamics is "permanently captured" in the higher-amplitude stable branch of the SIM* since the small range of force modulation does not allow it to reach the lower fold point and make the transition back to the lower-amplitude stable branch of the SIM (as happens in Fig. 3 where the force modulation is strong enough to enable such a high-to-low amplitude transition). It follows that the response of (9) due to force modulation $k_{II}(\varepsilon^2 t)$ becomes permanently "locked" into high-amplitude oscillations. Moreover, as shown in the next section this high-amplitude locking is independent of the initial conditions of (9).

One notes that maximum amplitude in k_I is equal to 3.7, which is slightly higher that the calculated value of k_+, and still the transition of the dynamics to the upper branch is not observed. This small discrepancy is not surprizing, since the accuracy of the primary averaging in the vicinity of the fold point may be substantially less accurate than $O(\varepsilon)$. More precise prediction would require quite formidable matching of singular asymptptic expansions in the vicinity of the fold point. At the moment, we leave it beyond the scope of this paper, since our intention is to primarily study the qualitative aspects of the dynamics due to modulated harmonic excitation, and, in doing so, highlight the benefits gained compared to the "classical" problem with no modulated harmonic excitation, i.e., system (1).

The corresponding results for the power input and the energy intake in (9) for excitations with modulated magnitudes $k_I(\varepsilon^2 t)$ and $k_{II}(\varepsilon^2 t)$ are depicted in Fig. 6. It is interesting to note the *drastic increase in input power and energy intake* (see Fig. 6b), due just a slight increase in the force modulation. This result clearly demonstrates the efficacy of predictively managing the energy intake in (9) by appropriately selecting the regions of the SIM that are tracked by the super-slowly modulated excitation. Conversely, one could achieve *drastic reduction of input power and energy intake* by imposing super-slow modulated forces that track only selected regions in the lower (linearized) region of the lower-amplitude stable branch of the SIM (this type of simulations is not shown here). Hence, the presented super-slow force modulation provides a versative design tool for predictively managing the energy input in nonlinear oscillators, through selective tracking of appropriate regions of the SIM.

In the next section, we numerically illustrate the main finding of this work, namely that *the described modulation-based methodology is indeed insensitive to the initial conditions of the system.* This result, which is due to the attracting nature of the SIM (i.e., its capacity to attract the fast dynamics), eliminates a major limitation of the current state-of-the-art in the area of harmonically forced nonlinear oscillators, which is related to the uncertainty of the steady state solution due to bi-stability, i.e., i.e., to the sensitivity of the forced dynamics to the initial conditions, in regions where multiple co-existing stable attractors exist.

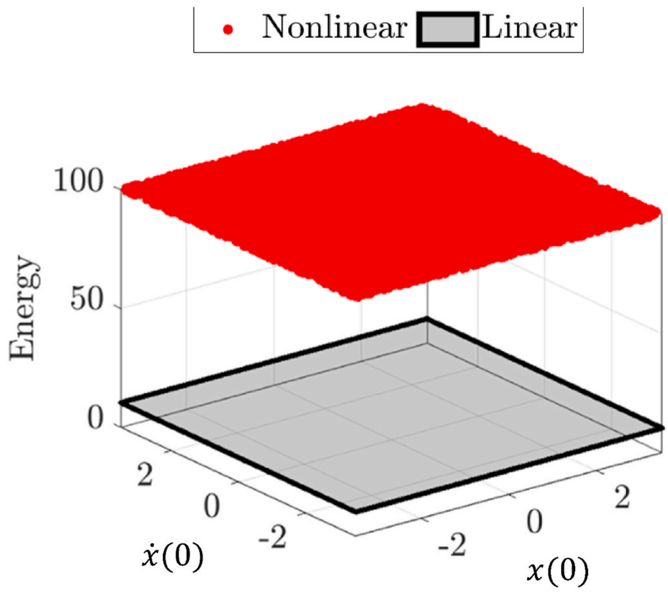


Fig. 7. Insensitivity of the response depicted in Fig. 2 (with corresponding energy intake shown in Fig. 3b) to the initial conditions.

3. Robustness study

By construction, the SIM attracts asymptotically the fast dynamics, i.e., the dynamics at the fast time scale t, as $t \gg 1$. This should render it invariant to changes in initial conditions, so that the results reported in the previous section should be insensitive to the specific choice of the initial conditions of (9). This represents robustness of the super-slow modulated results presented in Figs. 2–6. In this section, we perform a series of simulations to numerically confirm this result.

In Fig. 7, we examine the robustness of the super-slow, single cycle modulated response depicted in Fig. 2 (which was computed for zero initial conditions) to changes in initial conditions, by selecting 10^4 random pairs of initial conditions for the oscillator (9) in the range $x(0)$, $\dot{x}(0) \in [-2, 2]$, while keep the same modulated forcing magnitude (8), and the system parameters $\varepsilon = 0.01, \zeta = 0.1, \alpha = 1, \sigma = 2.0$. In each simulation, we evaluate the maximum energy intake, which for a response tracking the entire nonlinear hysteresis loop would reach the definitive extreme value shown in Fig. 3b (of ~ 100 normalized units). In the plot of Fig. 7, we depict the maximum energy intake for each of the randomly selected initial conditions for both nonlinear ($\alpha = 1$) and linear ($\alpha = 0$) cases, which confirm that the super-slow modulated response of Fig. 2 is independent of the initial conditions. It follows that bi-stability does not affect the forced dynamics, and a major uncertainty related to the choice of initial conditions is removed.

Next, we perform a similar study for the oscillator (9) subject to a multi-cycle modulated harmonic force with magnitude modulation $k_{II}(\varepsilon^2 t)$ in (11), and parameters $\varepsilon = 0.01, \zeta = 0.3, \alpha = 1.0$ and $\sigma = 2.0$. For zero initial conditions, the response of oscillator (9) is depicted in Fig. 5 and the corresponding power and energy intake in Fig. 6. In Fig. 8, we depict the maximum value of the energy intake that is reached at the end of the time window of the simulation ($t = 5 \times 10^4$ normalized time units) for each of the 10^4 random pairs of initial conditions considered. Contrary to the previous single cycle case, there are two types of responses that are possible in this case. One response type (similar to the response shown in Fig. 5 for zero initial conditions) involves *initial* attraction of the orbit by the lower-amplitude stable branch of the SIM, before the dynamics make a *permanent* transition to the larger-amplitude stable branch; the results indicate that this scenario is realized mainly for relatively small initial conditions. The second response type involves initial attraction directly by the larger-amplitude stable branch of the SIM, where the dynamics is captured permanently; the numerical simulation study indicates that this is the most plausible scenario for this multi-cycle modulated force excitation and yields permanently high values of power input right from the beginning of the dynamics. In fact, in both scenarios, the power inputs eventually attain the same permanently large values, the only difference being that in the first response type the transition into high power input is delayed by an initial, transient capture by the lower-amplitude stable branch of the SIM. It follows that again, insensitivity of the steady state forced dynamics is established in this case as well.

4. Concluding remarks

The well-known multiplicity of possible stable responses in (forced or unforced) nonlinear systems is perceived as one of most profound features of complexity. This work, somewhat paradoxically, demonstrates that multi-frequency quasiperiodic excitation may lead to less complex and fully predictable response regimes compared simple harmonic forcing excitation, by eliminating the sensitivity of the responses

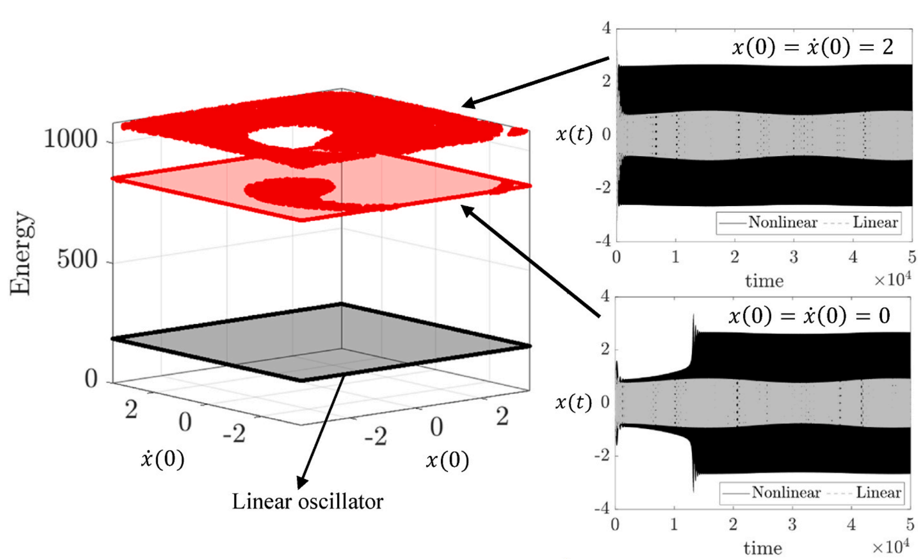


Fig. 8. Dependence of the response depicted in Fig. 4 (with corresponding energy intake shown in Fig. 6) to the initial conditions; a representative realization for each of the two types of responses is shown on the right.

to initial conditions, this eliminating the limitation of bi-stability. This finding, relevant for certain set of parameters and frequency-energy ranges, is a reminder that despite more than a century of active research, our understanding of forced responses even in the simplest models of nonlinear oscillators is still far from being complete and understood.

The presented results provide only an initial insight into a potentially very rich world of nonlinear responses of nonlinear systems subject to quasiperiodic excitations. Possible developments and applications include optimization (maximization or minimization) of the energy input with respect to the applied quasi-periodic excitations and the nonlinear system parameters, extensions for more complicated types of narrowband quasi-periodic excitations, as well as new suggestions for energy harvesting devices of drastically improved performance.

CRediT authorship contribution statement

Oleg V. Gendelman: Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Mohammad Bukhari:** Writing – review & editing, Validation, Software, Methodology, Investigation, Conceptualization. **Alexander F. Vakakis:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] A.H. Nayfeh, D. Mook, Nonlinear Oscillations, Wiley Interscience, New York, 1985.
[2] R.H. Rand, Lecture Notes on Nonlinear Vibrations, Cornell Library, 2012. https://hdl.handle.net/1813/28989.
[3] A.F. Vakakis, L.I. Manevitch, YuV. Mlkhlin, V.N. Pilipchuk, A.A. Zevin, Normal Modes and Localization in Nonlinear Systems, Wiley, 1996.
[4] G. Kerschen, M. Peeters, J.C. Golinval, A.F. Vakakis, Nonlinear normal modes, Part I: a useful framework for the structural dynamicist, Mech. Syst. Signal Process. 23 (1) (2009) 170–194.
[5] L.D. Landau, E.M. Lifshitz, Mechanics, Butterworth-Heinemann, 1976.
[6] V.I. Arnold, Mathematical Methods of Classical Mechanics, Springer, 1978.
[7] B.V. Chirikov, A universal instability of many-dimensional oscillator systems, Phys. Rep. 52 (5) (1979) 263–379.
[8] G.M. Zaslavsky, Hamiltonian Chaos and Fractional Dynamics, Oxford University Press, 2005.
[9] Y. Starosvetsky, O.V. Gendelman, Response regimes of linear oscillator coupled to nonlinear energy sink with harmonic forcing and frequency detuning, J. Sound Vib. 315 (2008) 746–765.
[10] O.V. Gendelman, Targeted energy transfer in systems with external and self – excitation, Proc. IME C J. Mech. Eng. Sci. 225 (2011) 2007–2043.
[11] F. Verhulst, Nonlinear Differential Equations and Dynamical Systems, Springer Verlag, Heidelberg and Berlin, 1990.
[12] L.I. Manevitch, Complex representation of dynamics of coupled nonlinear oscillators, in: L. Uvarova, A.E. Arinstein, A.V. Latyshev (Eds.), Mathematical Models of Nonlinear Excitations, Springer Verlag, Boston, 1999.
[13] O.V. Gendelman, A.F. Vakakis, L.A. Bergman, D.M. McFarland, Asymptotic analysis of passive nonlinear suppression of aeroelastic instabilities of a rigid wing in subsonic flow, SIAM J. Appl. Math. 70 (5) (2010) 1655–1677.